

**Sentiment Analysis of Rome Colosseum Visitor Reviews Using
Lexicon-Based and Machine Learning Approaches**

Submitted By:

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1. Introduction:

Sentiment analysis is one of the most common Natural Language Processing (NLP) tasks used to identify opinions and emotions expressed in text. Organizations use sentiment analysis to understand customer feedback, monitor public opinion, and improve products and services based on user experiences.

In this project, sentiment analysis was performed on visitor reviews of the Rome Colosseum. The reviews were classified into three sentiment categories: **Positive, Neutral, and Negative**. Two different approaches were explored during this project. The first approach used the VADER lexicon-based sentiment analyzer, while the second approach involved machine learning models trained using TF-IDF features.

Since the dataset contained a very large number of positive reviews compared to neutral and negative reviews, the project also focused on handling class imbalance using different techniques. Various machine learning models were trained and compared to identify the most suitable model for this dataset.

2. Problem Statement:

The main objective of this project was to build a sentiment classification system capable of predicting whether a visitor review expresses a positive, neutral, or negative opinion about the Rome Colosseum.

Several machine learning techniques were evaluated to determine which model performs best on an imbalanced dataset. Since relying only on overall accuracy can produce misleading results when one class dominates the dataset, additional evaluation metrics such as Macro Precision, Macro Recall, Macro F1-score, Balanced Accuracy, and Cohen's Kappa were used throughout the project.

The project also investigated different techniques for reducing the effect of class imbalance, including class weighting, random oversampling, and hyperparameter tuning using GridSearchCV.

3. Dataset Summary:

The dataset used in this project contains visitor reviews collected for the Rome Colosseum. Each review includes the review text along with its corresponding sentiment label. The reviews are categorized into three sentiment classes: Positive, Neutral, and Negative.

Initially, the dataset contained **8,285 reviews**. During preprocessing, one review became empty after text cleaning and was removed before training the machine learning models. As a result, **8,265 reviews** were used for model training and evaluation.

The sentiment distribution is highly imbalanced, with the majority of reviews belonging to the Positive class.

Dataset Information	Value
Total Reviews	8,285
Reviews Used for ML	8,265
Positive Reviews	7,453
Neutral Reviews	453
Negative Reviews	359
Number of Classes	3

The percentage distribution of each sentiment class is shown below.

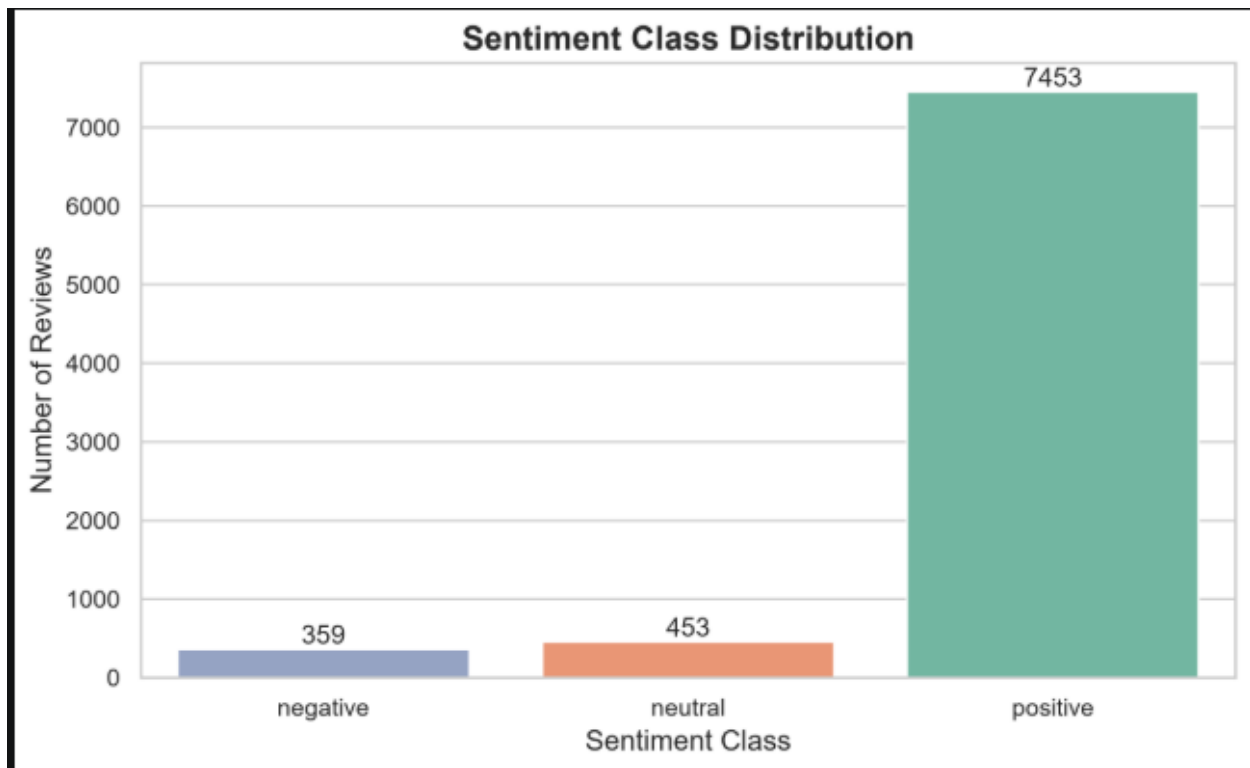
Sentiment	Percentage
Positive	90.18%
Neutral	5.48%
Negative	4.34%

These statistics clearly show that the dataset is highly imbalanced, making it difficult for machine learning models to correctly classify minority classes without applying imbalance-handling techniques.

4. Exploratory Data Analysis (EDA):

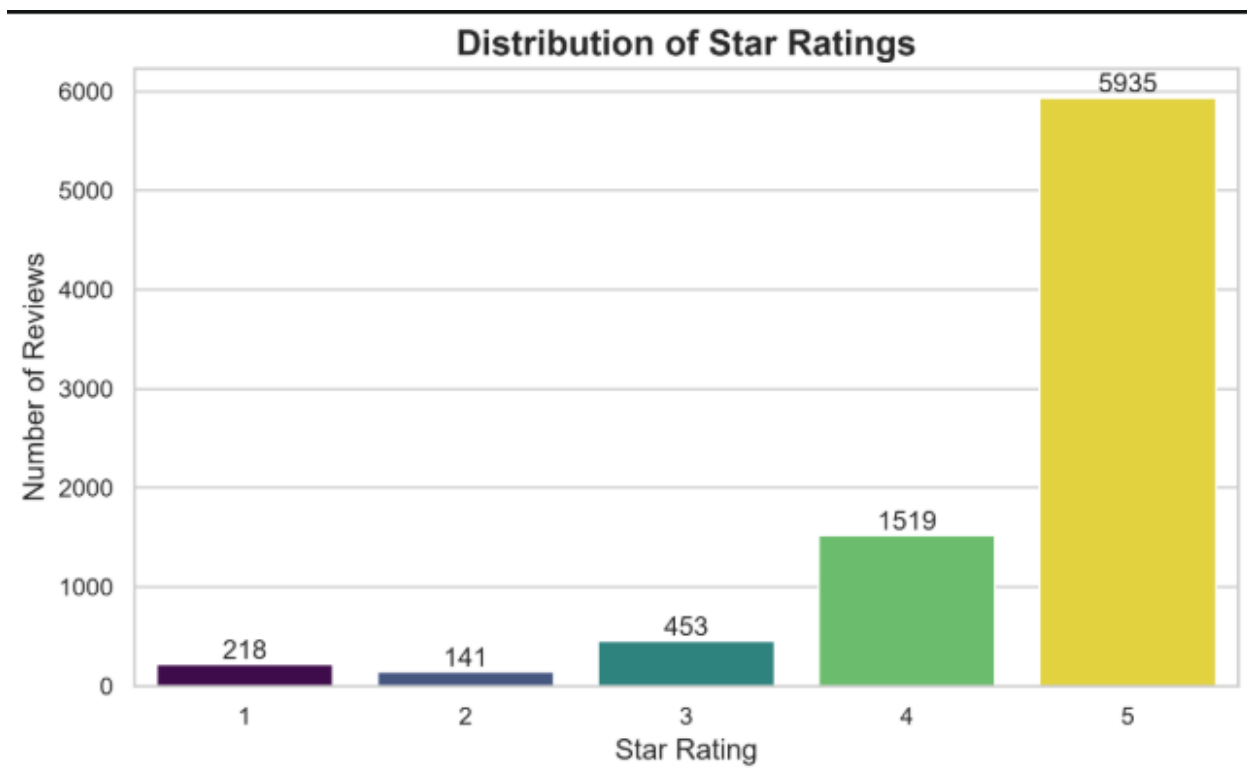
Before training any machine learning models, an exploratory data analysis (EDA) was performed to better understand the characteristics of the dataset. The analysis focused on identifying the sentiment distribution, rating patterns, and review lengths.

Figure 1:Sentiment class distribution Chart:



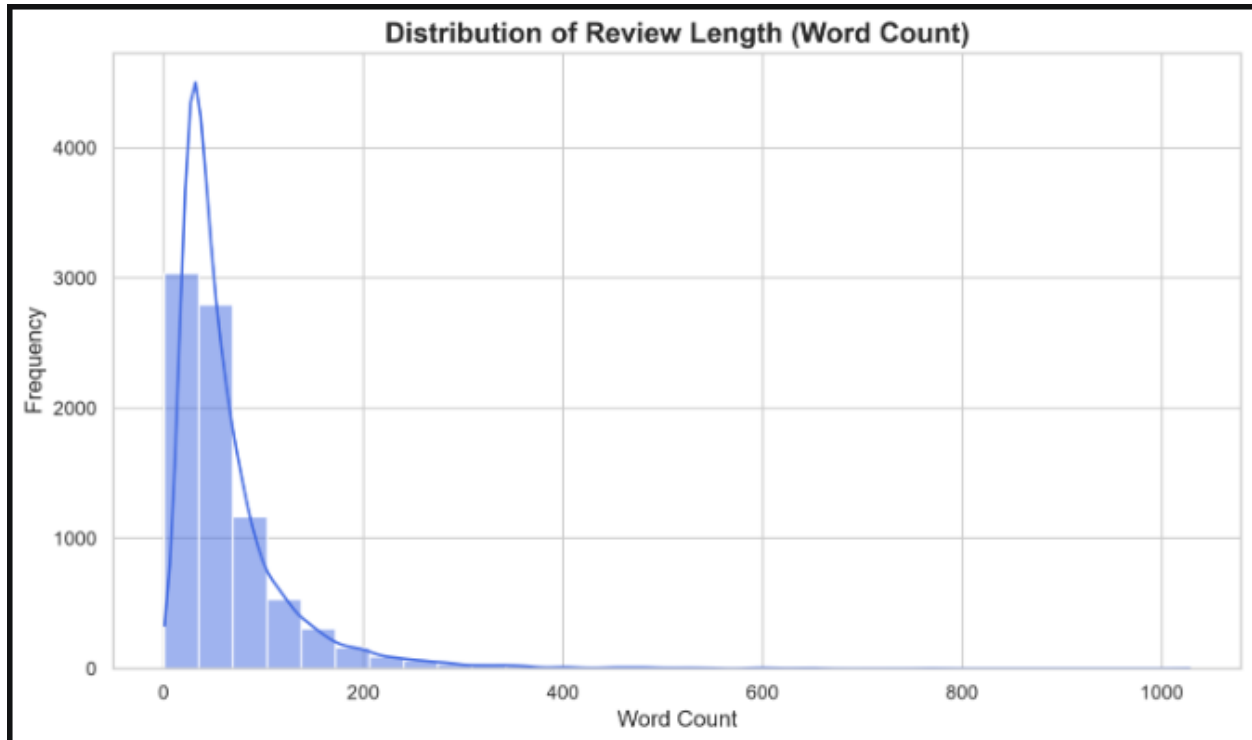
The first observation was that the majority of reviews were positive. Approximately 90% of all reviews expressed positive experiences, while only a small percentage belonged to the neutral and negative classes. This imbalance indicated that accuracy alone would not be sufficient for evaluating model performance.

Figure 2: Rating Distribution Chart:



The second analysis examined the distribution of review ratings. Most visitors assigned high ratings, particularly five-star ratings, which further confirmed the overwhelmingly positive nature of the dataset.

Figure 3: Review Length Distribution Chart:



The third analysis focused on review length. Review lengths varied significantly, with some users writing only a few words while others provided detailed descriptions of their experiences. This variation suggested that some reviews would provide much richer information for classification than others.

Overall, the exploratory data analysis revealed several important findings:

- The dataset is highly imbalanced.
- Positive reviews dominate the dataset.
- Most visitors assigned high ratings.
- Review lengths vary considerably across users.
- Class imbalance must be considered during model development and evaluation.

5. Text Preprocessing

Raw text data often contains unnecessary information such as punctuation, numbers, stop words, and inconsistent letter casing, which can negatively affect the performance of machine learning models. Therefore, a preprocessing pipeline was applied to clean and standardize the review text before feature extraction and model training.

The following preprocessing steps were performed on every review:

5.1 Lowercasing

All review text was converted into lowercase to ensure that words with different letter cases were treated as the same word. For example, "Amazing", "AMAZING", and "amazing" were all converted to "amazing".

5.2 Removing Punctuation and Special Characters

Punctuation marks and special symbols such as commas, periods, quotation marks, brackets, and other non-alphabetic characters were removed because they do not contribute significantly to sentiment classification.

5.3 Removing Numbers

Numerical values were removed from the reviews since they generally do not provide meaningful information for predicting sentiment.

5.4 Tokenization

Each review was divided into individual words (tokens). Tokenization allows the text to be processed word by word, making it easier to perform further cleaning operations.

5.5 Stop Word Removal

Common English stop words such as *the*, *is*, *are*, *of*, *and*, and *to* were removed because they appear frequently but carry little semantic information for sentiment analysis.

However, important negation words such as "**not**", "**no**", and "**never**" were intentionally preserved because removing them could completely change the meaning of a sentence. For example, "not good" should remain different from "good".

5.6 Lemmatization

Lemmatization was applied to convert words into their base forms while preserving their actual meanings. For example:

- *visited* → *visit*
- *running* → *run*
- *better* → *good* (where applicable)

This helped reduce vocabulary size while keeping the semantic meaning of words intact.

5.7 TF-IDF Feature Extraction

After preprocessing, the cleaned reviews were converted into numerical features using **Term Frequency–Inverse Document Frequency (TF-IDF)** vectorization.

A maximum vocabulary size of **5,000 features** was selected to represent the most informative words in the dataset while reducing computational complexity.

The final TF-IDF matrices had the following dimensions:

Dataset	Shape
Training Set	(6612, 5000)
Testing Set	(1653, 5000)

The data was divided using an **80:20 train-test split** with stratification to preserve the original class distribution in both training and testing datasets.

6. Machine Learning Models and Evaluation

Several machine learning models were trained and evaluated to determine the most suitable approach for sentiment classification. Since the dataset was highly imbalanced, multiple techniques were explored to improve the prediction of minority classes.

The following models were implemented during this project:

- Logistic Regression (Baseline)
- Multinomial Naïve Bayes
- Logistic Regression with Class Weighting
- Logistic Regression with Random Oversampling
- Logistic Regression with Hyperparameter Tuning (GridSearchCV)

To ensure a fair comparison, all models were trained using the same TF-IDF features and evaluated on the same testing dataset.

Instead of relying only on overall accuracy, additional evaluation metrics were used because the dataset contained approximately **90% positive reviews**. Accuracy alone could produce misleading results by favoring the majority class.

The primary evaluation metric selected for this project was **Macro F1-score**, as it gives equal importance to all sentiment classes regardless of their size. Other evaluation metrics included:

- Accuracy

- Macro Precision
- Macro Recall
- Balanced Accuracy
- Cohen's Kappa
- Confusion Matrix
- Classification Report

Using these metrics provided a more reliable comparison of model performance, particularly for the minority Neutral and Negative classes.

7. Model Comparison

Several machine learning models were trained and evaluated to determine the most effective approach for sentiment classification. Each model was trained using the same TF-IDF feature representation and evaluated on the testing dataset using identical evaluation metrics.

Since the dataset is highly imbalanced, **Macro F1-score** was selected as the primary evaluation metric because it gives equal importance to Positive, Neutral, and Negative sentiment classes. Accuracy was also reported, but it was not considered the primary criterion for selecting the best model.

Table 1 summarizes the performance of all evaluated models.

Table 1: Performance Comparison of Machine Learning Models

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Balanced Accuracy	Cohen's Kappa
Logistic Regression (Baseline)	0.9087	0.6923	0.3879	0.4120	0.3879	0.1453
Multinomial Naïve Bayes	0.9014	0.3005	0.3333	0.3160	0.3333	0.0000
Logistic Regression (Class-Weighted)	0.8518	0.5384	0.5984	0.5569	0.5984	0.3647
Random Oversampling + Logistic Regression	0.8615	0.5330	0.5641	0.5446	0.5641	0.3474
GridSearchCV Logistic Regression	0.9074	0.6375	0.4615	0.5076	0.4615	0.3020

The results show that the **baseline Logistic Regression** achieved the highest overall accuracy of **90.87%**, but its Macro F1-score was only **0.4120**. This indicates that although the model performed well on the majority Positive class, it struggled to correctly classify Neutral and Negative reviews.

The **Multinomial Naïve Bayes** model produced the weakest performance among all evaluated models. It classified almost all reviews as Positive, resulting in a Macro F1-score of only **0.3160**. This demonstrates that Naïve Bayes was not suitable for this highly imbalanced dataset.

Applying **class weighting** significantly improved the model's ability to recognize minority classes. Although the overall accuracy decreased to **85.18%**, the Macro F1-score increased to **0.5569**, which was the highest among all evaluated models. The Balanced Accuracy and Cohen's Kappa values also improved considerably, indicating more balanced predictions across all sentiment classes.

The **Random Oversampling** technique also improved minority class recognition. It achieved a Macro F1-score of **0.5446**, which is slightly lower than the class-weighted Logistic Regression but substantially better than the baseline model.

Finally, **GridSearchCV** was used to optimize the Logistic Regression hyperparameters. The tuned model achieved a Macro F1-score of **0.5076**, which was an improvement over the baseline Logistic Regression but did not outperform the class-weighted model.

Overall, the results indicate that handling class imbalance is more beneficial than simply maximizing overall accuracy.

8. Class Imbalance Analysis

One of the main challenges encountered during this project was the highly imbalanced class distribution. Approximately **90.18%** of all reviews belonged to the Positive class, while only **5.48%** were Neutral and **4.34%** were Negative.

Such an imbalanced dataset can lead machine learning models to become biased toward the majority class. A classifier may achieve very high accuracy simply by predicting most reviews as Positive while failing to correctly identify Neutral and Negative reviews.

For this reason, relying solely on accuracy would have produced misleading conclusions. Instead, Macro Precision, Macro Recall, Macro F1-score, Balanced Accuracy, and Cohen's Kappa were used to provide a more reliable evaluation of model performance.

To reduce the impact of class imbalance, three different techniques were investigated:

1. Baseline Logistic Regression

The baseline Logistic Regression model was trained without applying any imbalance handling technique. Although it achieved a high accuracy of **90.87%**, its Macro F1-score remained relatively low because the model favored the majority Positive class.

2. Class Weighting

The second approach used **class_weight='balanced'** within Logistic Regression. This technique automatically assigned greater importance to the minority classes during training.

This approach produced the highest Macro F1-score (**0.5569**) among all evaluated models. The improvement in Balanced Accuracy and Cohen's Kappa also indicated that the classifier became much more effective at recognizing Neutral and Negative reviews.

3. Random Oversampling

Random Oversampling duplicated minority class samples until all classes contained an equal number of training examples.

This technique improved minority class prediction and achieved a Macro F1-score of **0.5446**. However, because duplicated samples were introduced, there is a possibility of overfitting during training.

4. Hyperparameter Tuning

GridSearchCV was used to search for the best combination of Logistic Regression hyperparameters using five-fold cross-validation.

The best parameters selected were:

- **C = 10**
- **Solver = lbfgs**

Although hyperparameter tuning improved the baseline model, it still produced a lower Macro F1-score than the class-weighted Logistic Regression.

Based on the experimental results, **Logistic Regression with class weighting** was selected as the final model because it achieved the best balance between overall performance and minority class recognition.

9. Error Analysis

To better understand the limitations of the selected model, an error analysis was performed using the predictions generated by the best-performing model, **Logistic Regression with Class Weighting**. A total of **245 reviews** were incorrectly classified out of **1,653 testing reviews**, resulting in a misclassification rate of approximately **14.82%**.

A sample of the misclassified reviews was examined to identify common patterns and understand why the model made incorrect predictions.

The first observation was that many reviews contained **mixed sentiments**. Several visitors expressed both positive and negative opinions within the same review. For example, a review might describe the Colosseum as an amazing historical site while also complaining about long queues, overcrowding, or expensive tickets. Since both positive and negative words appear together, the classifier sometimes struggled to determine the overall sentiment.

The second common pattern involved **neutral reviews**. Many neutral reviews were descriptive rather than opinion-based. They simply explained facts about the visit without expressing strong positive or negative emotions. Because these reviews contained only a few sentiment-related words, the model often predicted them as Positive or Negative instead of Neutral.

Another challenge was found in **short reviews**. Reviews consisting of only a few words provided very limited contextual information for the TF-IDF model. Simple comments such as *"Nice place"* or *"Worth visiting"* contained too few informative features, making accurate classification more difficult.

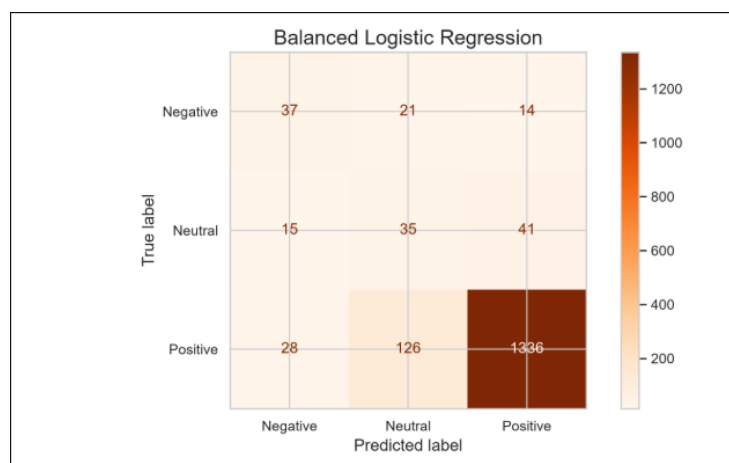
The analysis also showed that reviews mentioning **queues, ticket prices, crowds, waiting time, or hot weather** frequently caused prediction errors. Although many visitors ultimately enjoyed their experience, the presence of complaint-related words sometimes caused the model to assign a Negative sentiment even when the overall review was Positive.

Although negation words such as **"not"**, **"no"**, and **"never"** were preserved during preprocessing, the model still experienced difficulty interpreting complex sentence structures involving negation. For example, phrases such as *"not as crowded as expected"* or *"not bad at all"* require contextual understanding that traditional TF-IDF models cannot fully capture.

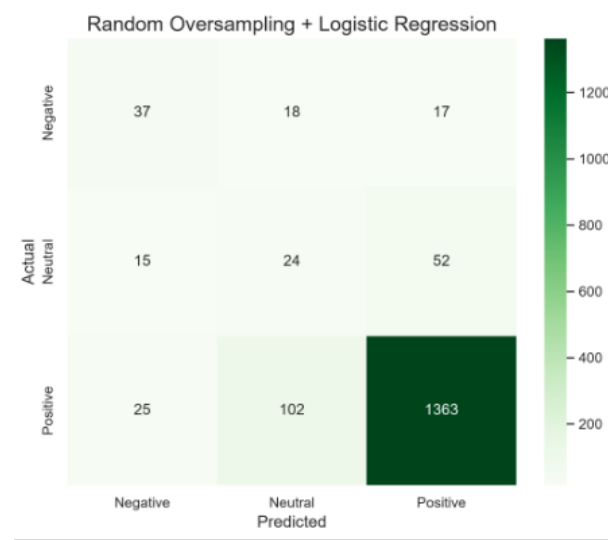
Overall, the error analysis indicates that most misclassifications occurred because the reviews contained multiple opinions, subtle language, or insufficient context. Since TF-IDF represents documents using word frequencies rather than sentence meaning, it cannot always capture the true intent of the reviewer.

10. TWO best models with confusion matrix:

First Best Model: Logistic Regression With Class weighting:



Second Best Model: Logistic Regression With Random OverSampling:



11. Conclusion

The objective of this project was to develop a sentiment analysis system capable of classifying Rome Colosseum visitor reviews into Positive, Neutral, and Negative categories. Throughout the project, both lexicon-based and machine learning approaches were implemented and evaluated.

The VADER sentiment analyzer provided a useful baseline for comparison and achieved good overall accuracy. However, its performance on the minority sentiment classes was limited due to the highly imbalanced nature of the dataset.

Several machine learning models were then trained using TF-IDF features, including Logistic Regression, Multinomial Naïve Bayes, Logistic Regression with class weighting, Random Oversampling, and GridSearchCV. Each model was evaluated using Macro Precision, Macro Recall, Macro F1-score, Balanced Accuracy, Cohen's Kappa, Confusion Matrix, and Classification Report.

Among all evaluated approaches, **Logistic Regression with Class Weighting** achieved the best overall performance, obtaining a **Macro F1-score of 0.5569**. Although its overall accuracy (**85.18%**) was lower than the baseline Logistic Regression, it produced a much better balance across all three sentiment classes. Since Macro F1-score gives equal importance to every class regardless of dataset size, this model was selected as the most appropriate solution for the imbalanced sentiment classification task.

The experiments also demonstrated that handling class imbalance significantly improves the recognition of minority classes. Both class weighting and random oversampling produced

better Macro F1-scores than the baseline model, highlighting the importance of selecting suitable imbalance-handling techniques when working with real-world datasets.

12. Limitation

One limitation of this study is the highly imbalanced nature of the dataset, where approximately **90% of the reviews are positive**. Even after applying class weighting and oversampling techniques, the models continued to struggle with correctly identifying Neutral and Negative reviews. In addition, the TF-IDF representation relies on word frequencies and cannot fully understand sentence context, sarcasm, or mixed opinions expressed within a review.

13. Future Work

Several improvements can be made to further enhance the performance of the sentiment classification system. Instead of TF-IDF features, transformer-based language models such as **BERT**, **RoBERTa**, or **DistilBERT** could be used to better capture contextual meaning and semantic relationships within the text. Advanced imbalance-handling techniques, such as **SMOTE** or data augmentation, could also be explored to improve minority-class prediction. Furthermore, experimenting with ensemble learning methods and larger datasets may lead to even better classification performance.